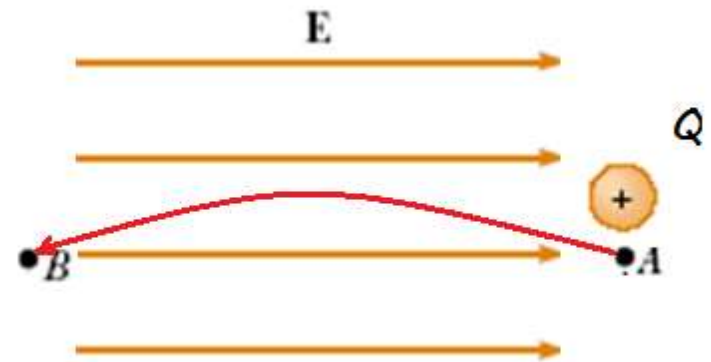
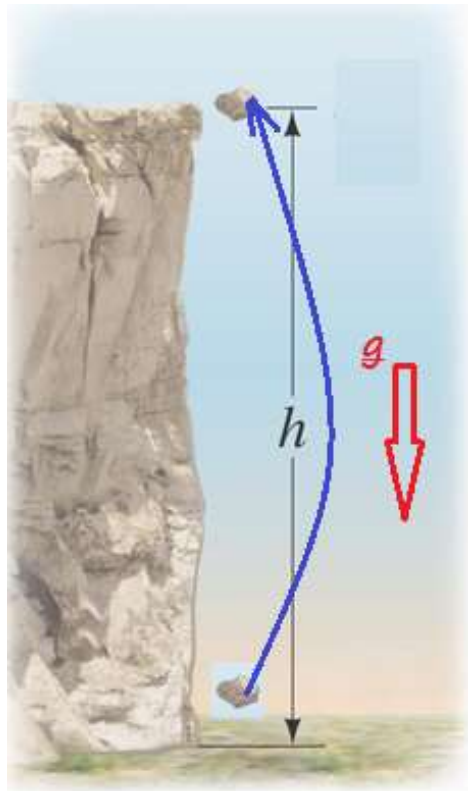


Electric Potential

Prepared by:

Dr. Yasser El-Batawy

Electric Potential



Electric Potential

- The potential difference $V_B - V_A$ between two points, A and B, in a region containing an electrostatic field is defined as the work done against the electrostatic force per unit test charge during its motion from the initial point A to the final point B. That is

$$V_B - V_A = -\frac{W_{AB}}{q_o} = -\frac{1}{q_o} \int_A^B \mathbf{F}_e \cdot d\mathbf{l} \quad \text{But } \frac{\mathbf{F}_e}{q_o} = \mathbf{E}$$

$$\therefore V_B - V_A = -\int_A^B \mathbf{E} \cdot d\mathbf{l}$$

Units of V is J/C = Volt

Electric Potential

Notice the following:

- The potential difference is independent of the test charge. It depends only on the sources and the location of the two points relative to them.
- No path is specified for the integration since it is independent of the path.
- The physical quantity, which can be measured in the lab is the potential difference and not the potential.

$$V_B - V_A = - \int_A^B \mathbf{E} \cdot d\mathbf{l}$$

Electric Potential of a point

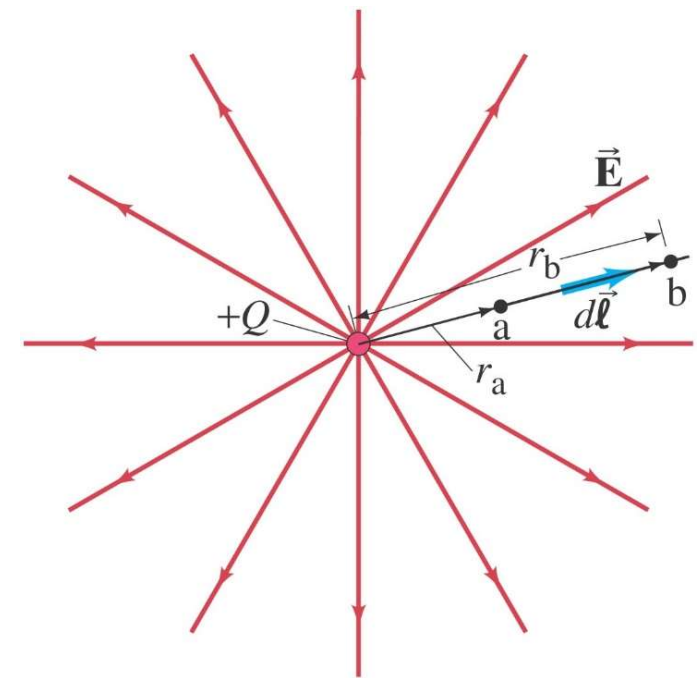
- ▶ one can choose a convenient reference where the potential is arbitrarily given the value zero.
- ▶ $V_{\infty} = 0$
- ▶ Then , the electric potential at a certain point, B can be expressed as

$$V_B = - \int_{\infty}^B E \cdot dl$$

Electric Potential due to Point Charge

$$\begin{aligned} V_b - V_a &= - \int_{r_a}^{r_b} \vec{\mathbf{E}} \cdot d\vec{\ell} = - \frac{Q}{4\pi\epsilon_0} \int_{r_a}^{r_b} \frac{1}{r^2} dr \\ &= \frac{1}{4\pi\epsilon_0} \left(\frac{Q}{r_b} - \frac{Q}{r_a} \right) \end{aligned}$$

$$\lim_{r_A \rightarrow \infty} V_A = 0, \text{ then } V = k_e \frac{q}{r}$$



V for Point Charge system

$$V_B - V_A = k_e \sum_{m=1}^n q_m \left(\frac{1}{r_{mB}} - \frac{1}{r_{mA}} \right)$$

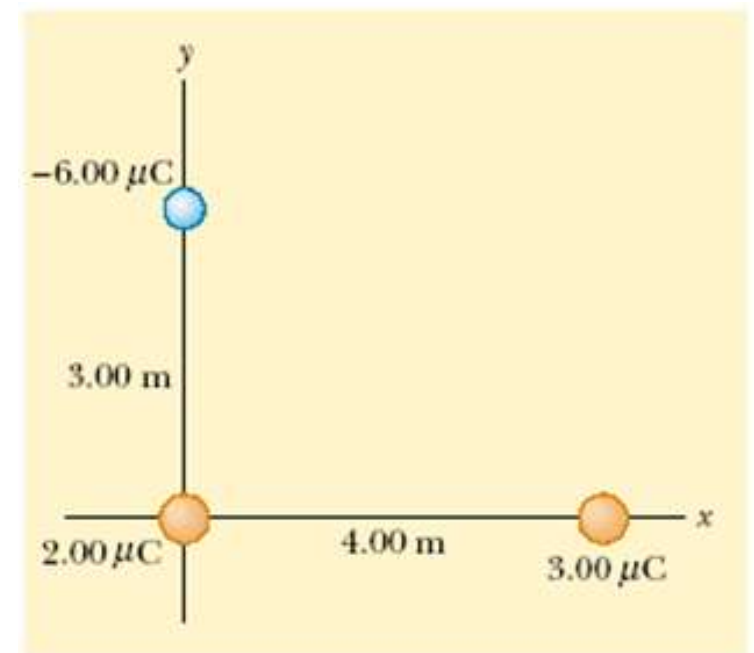
$$V = k_e \sum_{m=1}^n \frac{q_m}{r_m}$$

Example 1 :

A charge $q_1 = 2 \mu\text{C}$ is located at the origin, and a charge $q_2 = -6 \mu\text{C}$ is located at $(0, 3.00) \text{ m}$, as shown in figure.

(a) Find the total electric potential due to these charges at the point P , whose coordinates are $(4.00, 0) \text{ m}$.

(b) Find the change in potential energy of a $3.00\text{-}\mu\text{C}$ charge as it moves from infinity to point P .



Solution

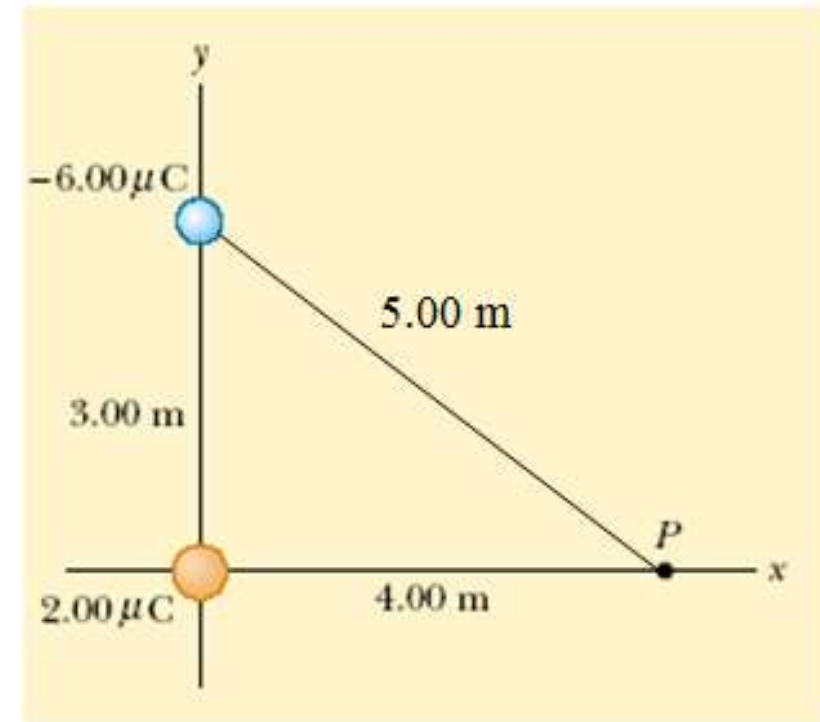
$$(a) V_P = k_e \left(\frac{q_1}{r_1} + \frac{q_2}{r_2} \right)$$

$$\therefore V_P = 9 \times 10^9 \left(\frac{2.00 \times 10^{-6}}{4.00} + \frac{-6.00 \times 10^{-6}}{5.00} \right)$$

$$\therefore V_P = -6.300 \times 10^3 \text{ V}$$

$$(b) \Delta U = q(V_{\text{final}} - V_{\text{initial}})$$

$$\therefore \Delta U = 3.00 \times 10^{-6} (-6.30 \times 10^3 - 0) = -1.89 \times 10^{-2} \text{ J}$$



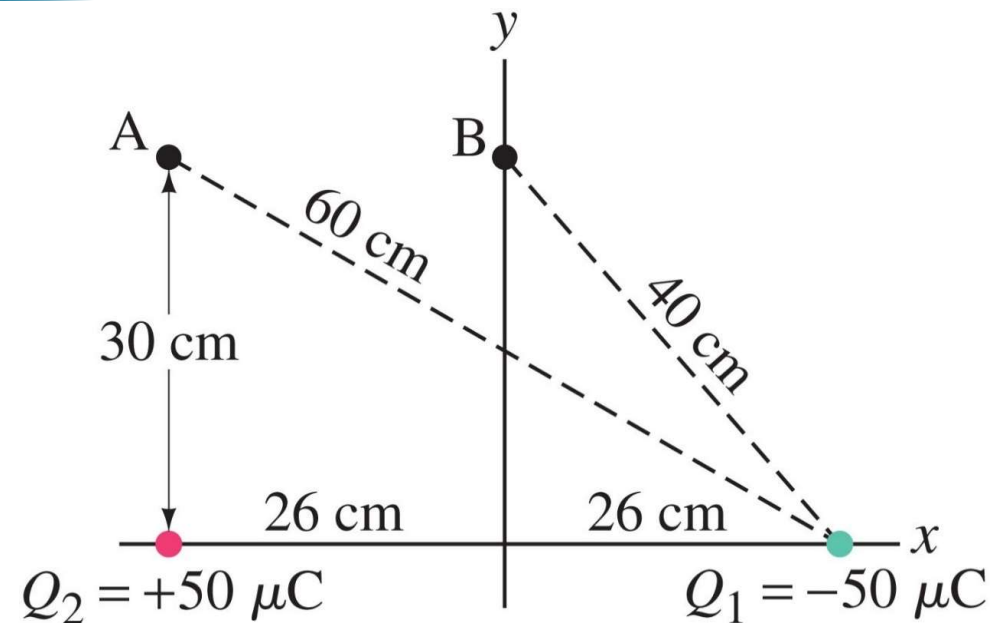
Example 2 :

- ▶ Work required to bring two positive charges close together.
- ▶ What minimum work must be done by an external force to bring a charge $q = 3.00 \mu\text{C}$ from a great distance away (take $r = \infty$) to a point 0.500 m from a charge $Q = 20.0 \mu\text{C}$?

Answer: $W = 1.08 \text{ J}$

Example 3:

- Calculate the electric potential (a) at point A in the figure due to the two charges shown, and (b) at point B.



Answer : $V_a = 7.5 \times 10^5 \text{ V}$ & $V_b = 0$

V for Gauss' applications

- ▶ Determine **E** using Gauss' law in the regions of interest

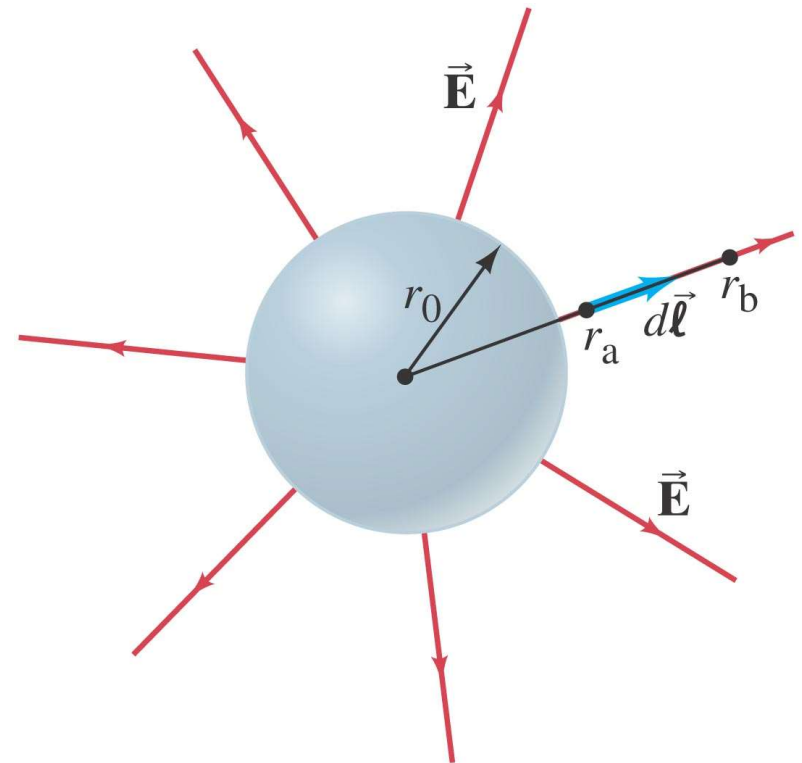
$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{encl}}}{\epsilon_0}$$

- ▶ Determine V using the general relation

$$V = - \int_{\infty}^r E \cdot dl$$

Example 5:

- Determine the potential at a distance r from the center of a uniformly charged spherical surface of radius r_0 for (a) $r > r_0$, (b) $r < r_0$. The total charge on the sphere is Q .



Solution

► For $r < r_o$

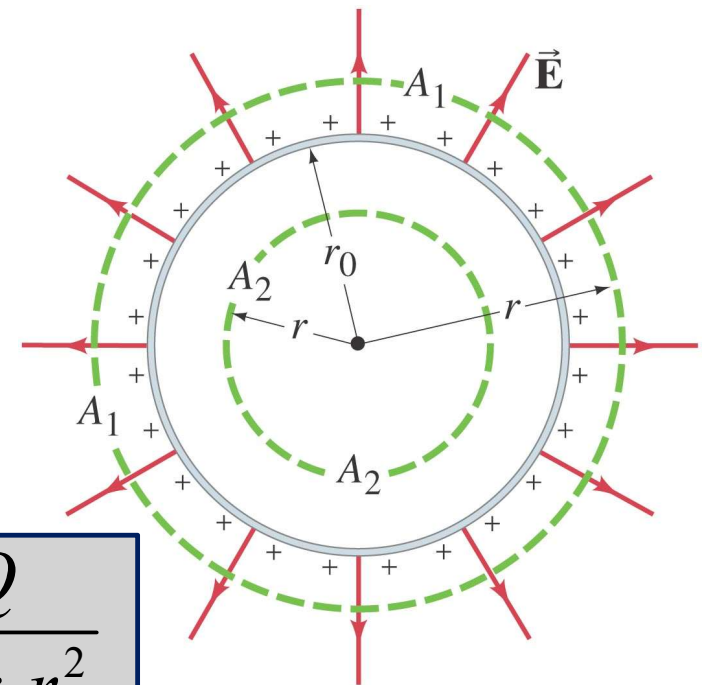
$$\Phi_E = \oint \mathbf{E} \cdot d\mathbf{A} = \frac{q_{in}}{\epsilon_o} = 0$$

$$\mathbf{E} = 0$$

► For $r > r_o$

$$\Phi_E = \oint \mathbf{E} \cdot d\mathbf{A} = \oint E \cos 0^\circ dA = \oint E dA = \frac{q_{in}}{\epsilon_o}$$

$$\therefore E = \frac{Q}{4\pi\epsilon_o r^2}$$



Solution

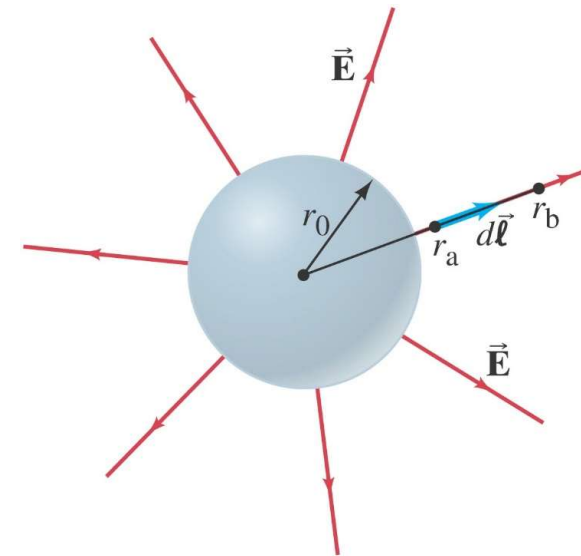
► By applying

$$V = -\int_{\infty}^r \mathbf{E} \cdot d\mathbf{l} = -\int_{\infty}^r E dr$$

► Then,

► a. $V = Q/4\pi\epsilon_0 r$ for $r > r_o$

► b. $V = Q/4\pi\epsilon_0 r_o$ for $r < r_o$



Example 6:

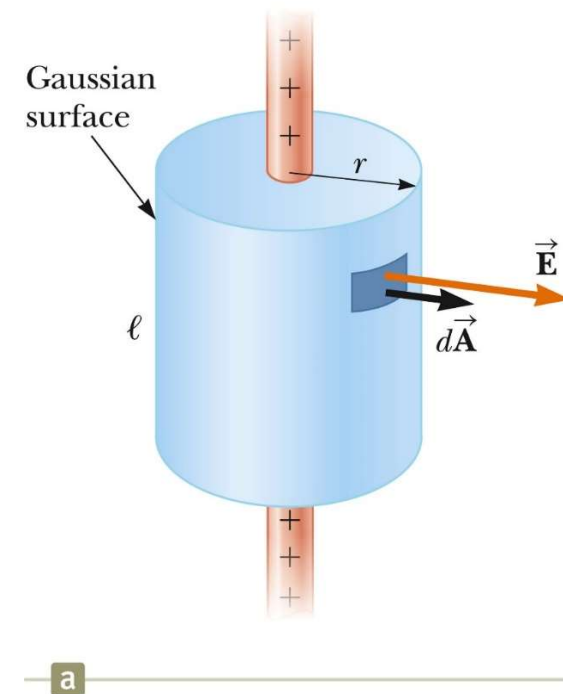
- ▶ From Gauss's law, the electric field set up by a uniform line of charge is pointing radially away from the line and λ is the linear charge density along the line. Derive an expression for the potential difference between $r = r_1$ and $r = r_2$.

Solution

$$\mathbf{E} = \frac{\lambda}{2\pi\epsilon_0 r} \hat{\mathbf{r}}$$

$$V_2 - V_1 = - \int_{r_1}^{r_2} \mathbf{E} \cdot d\mathbf{r} = - \int_{r_1}^{r_2} \frac{\lambda}{2\pi\epsilon_0 r} dr$$

$$V_2 - V_1 = \left[\frac{-\lambda}{2\pi\epsilon_0} \ln\left(\frac{r_2}{r_1}\right) \right]$$



Electric Field from V function

$$dV = -\mathbf{E} \cdot d\mathbf{l}$$

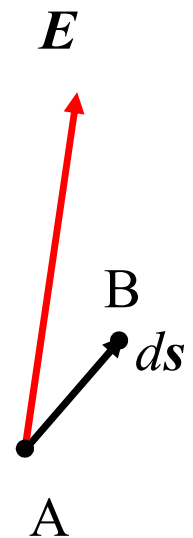
The vector field can be written as $\mathbf{E} = E_x \hat{\mathbf{i}} + E_y \hat{\mathbf{j}} + E_z \hat{\mathbf{k}}$

$$\text{and } d\mathbf{l} = dx \hat{\mathbf{i}} + dy \hat{\mathbf{j}} + dz \hat{\mathbf{k}}$$

$$\therefore dV = -(E_x dx + E_y dy + E_z dz)$$

$$\therefore E_x = - \left. \frac{dV}{dx} \right|_{y \text{ and } z = \text{constant}}$$

$$\therefore E_x = -\frac{\partial V}{\partial x}, E_y = -\frac{\partial V}{\partial y}, E_z = -\frac{\partial V}{\partial z}$$



Electric Field from V function

$$\therefore \mathbf{E} = -\nabla V$$

$$\mathbf{E} = -\left(\frac{\partial V}{\partial x} \hat{\mathbf{i}} + \frac{\partial V}{\partial y} \hat{\mathbf{j}} + \frac{\partial V}{\partial z} \hat{\mathbf{k}}\right)$$

where the vector operator

$$\nabla = \left(\hat{\mathbf{i}} \frac{\partial}{\partial x} + \hat{\mathbf{j}} \frac{\partial}{\partial y} + \hat{\mathbf{k}} \frac{\partial}{\partial z}\right)$$

∇V is called the gradient of V .

Example 7:

- Over a certain region of space, the electric potential is $V = 5x - 3x^2y + 2yz^2$. Find the expressions for the x , y , and z components of the electric field over this region. What is the magnitude of the field at the point P that has coordinates $(1, 0, -2)$ m?

Solution

$$E_x = -\frac{\partial V}{\partial x} = \boxed{-5 + 6xy} = -5 + 6(1)(0) = -5$$

$$E_y = -\frac{\partial V}{\partial y} = \boxed{+3x^2 - 2z^2} = 3(1)^2 - 2(-2)^2 = -5$$

$$E_z = -\frac{\partial V}{\partial z} = \boxed{-4yz} = -4(0)(-2) = 0$$

$$E = \sqrt{E_x^2 + E_y^2 + E_z^2} = \sqrt{(-5)^2 + (-5)^2 + 0^2} = \boxed{7.07 \text{ N/C}}$$

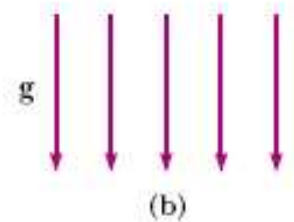
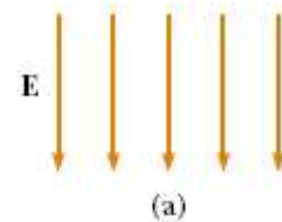
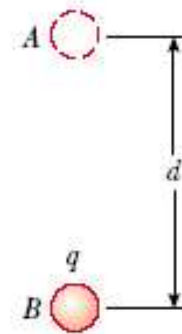
Electric Potential Energy, U

- ▶ Comparing the expressions for ΔU and ΔV , the relation between the difference in potential between two points and the change in potential energy resulting from changing the location of a test charge is:

$$\Delta V = \frac{\Delta U}{q_o}$$

- ▶ If a charge of q_o moves from A to B in a region, the change of the electric potential energy is

$$\Delta U = U_B - U_A = q_o (V_B - V_A)$$



Check

<https://www.youtube.com/watch?v=gnUszuDVqck>

<https://www.youtube.com/watch?v=IftlbXZjqBM>

<https://www.youtube.com/watch?v=kX-OZrfV-Eg>

<https://www.youtube.com/watch?v=f59TN2EetJk>